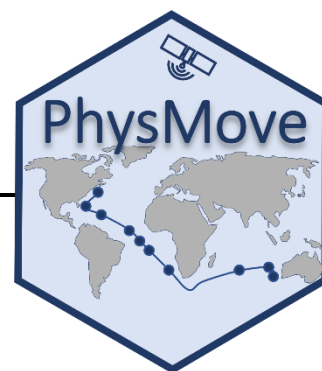


PhysMove: Introduction

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PhysMove contains a comprehensive collection of methods for documenting species' movement and space-use patterns from satellite telemetry data. This vignette demonstrates how to calculate each of the methods and reviews all relevant functions and parameters. We demonstrate each function with a simulated telemetry dataset, called tracks, which is automatically loaded with PhysMove ([Explore tracks dataset](#)). For further details on our methods and interpreting results please see our corresponding manuscript.

Installation

We recommend users install the development version of PhysMove from GitHub using the devtools R package. Note that the authentication token included below will only be required while the accompanying manuscript is under review, PhysMove will be open access on GitHub once the manuscript is accepted for publication.

```
# Install the devtools package from CRAN (if required)
install.packages("devtools")

# Download the development version from GitHub:
devtools::install_github("HannahCalich/PhysMove", auth_token =
"ghp_6UF7PMT6Fg8w2Lq71RtBbRvQVfk7pX2CEatC", build_vignettes = TRUE)

# Load PhysMove
library(PhysMove)
```

Data formatting

PhysMove was designed to be user-friendly and most functions only require you to input a data frame containing standard telemetry data. The input data frame must only contain these four columns in the following order: *ref*, *lon*, *lat*, and *day*.

Columns must be formatted as follows:

- *ref*: the unique telemetry tag ID number for each animal in numeric format (note that characters are not accepted because they can be slower to process than integers, so please convert all reference IDs to integers before proceeding)
- *lon* and *lat*: the longitude (-180 to + 180) and latitude (-90 to +90) in decimal degrees of each position estimate, respectively, in numeric format, and
- *day*: the datetime stamp for each location estimate in POSIXct format following yyyy-mm-dd hh:mm:ss.

The CheckTracks function can be used to confirm your input data are formatted as described above. This function checks that column names are in order and that each column is in the correct format as described above. Note that this function does not evaluate data quality or quantity.

```
# Check your data are formatted correctly  
CheckTracks(data) # replace "data" with your data frame
```

You can also compare your data frame to our sample dataset tracks to ensure your data are formatted correctly.

Explore tracks dataset

tracks is a simulated telemetry dataset of 25 unique tracks with a defined set of movement parameters that was designed to demonstrate each of the PhysMove functions. In detail, Tracks was created using a biased, uncorrelated random walk model with variable step lengths drawn from an exponential distribution with $\lambda = 0.125$. We defined the turning angles such that 30% indicated directed forward movement (movements with angles $<30^\circ$ or $>330^\circ$), and 30% indicated directed return movement (angles between 150° - 210°), allowing the remainder (40%) to be randomly between 0° - 360° . These metrics were chosen because they are broadly consistent with literature interpretations of animals moving through a resource-rich habitat. See corresponded manuscript for further details. The code used to make the tracks dataset is available in the PhysMove doc folder on GitHub as "CreateTracks.R".

```
# Preview the first 6 rows of the tracks dataset  
head(tracks)  
#>   ref      lon      lat      day  
#> 1   1 0.5310173 0.5310173 2017-10-13 15:00:00  
#> 2   1 0.5156939 0.5500691 2017-10-14 15:00:00  
#> 3   1 0.5052581 0.5158941 2017-10-15 15:00:00  
#> 4   1 0.5247597 0.4555179 2017-10-16 15:00:00  
#> 5   1 0.5491573 0.2831650 2017-10-17 15:00:00  
#> 6   1 0.5670918 0.2875133 2017-10-18 15:00:00  
  
# Determine the structure of the tracks dataset  
str(tracks)  
#> 'data.frame':   15623 obs. of  4 variables:
```

```
#> $ ref: num 1 1 1 1 1 1 1 1 1 1 ...
#> $ lon: num 0.531 0.516 0.505 0.525 0.549 ...
#> $ lat: num 0.531 0.55 0.516 0.456 0.283 ...
#> $ day: POSIXct, format: "2017-10-13 15:00:00" "2017-10-14 15:00:00"
"2017-10-15 15:00:00" "2017-10-16 15:00:00" ...
```

Create a map of the tracks dataset

A basic map of your telemetry data can be created using our `PlotTracks()` function (Figure V1).

`PlotTracks()` requires a data frame with telemetry data (as described above) and includes three optional parameters:

- `ref`: plot specific tracks based on their reference IDs (`ref=NULL`, by default),
- `tracks`: connect points with lines (`tracks=TRUE`, by default), and
- `colours`: edit the colours used in the map (`colours=rainbow`, by default).

`PlotTracks(tracks)`

```
#> Note: World polygon does not overlap with location data
```

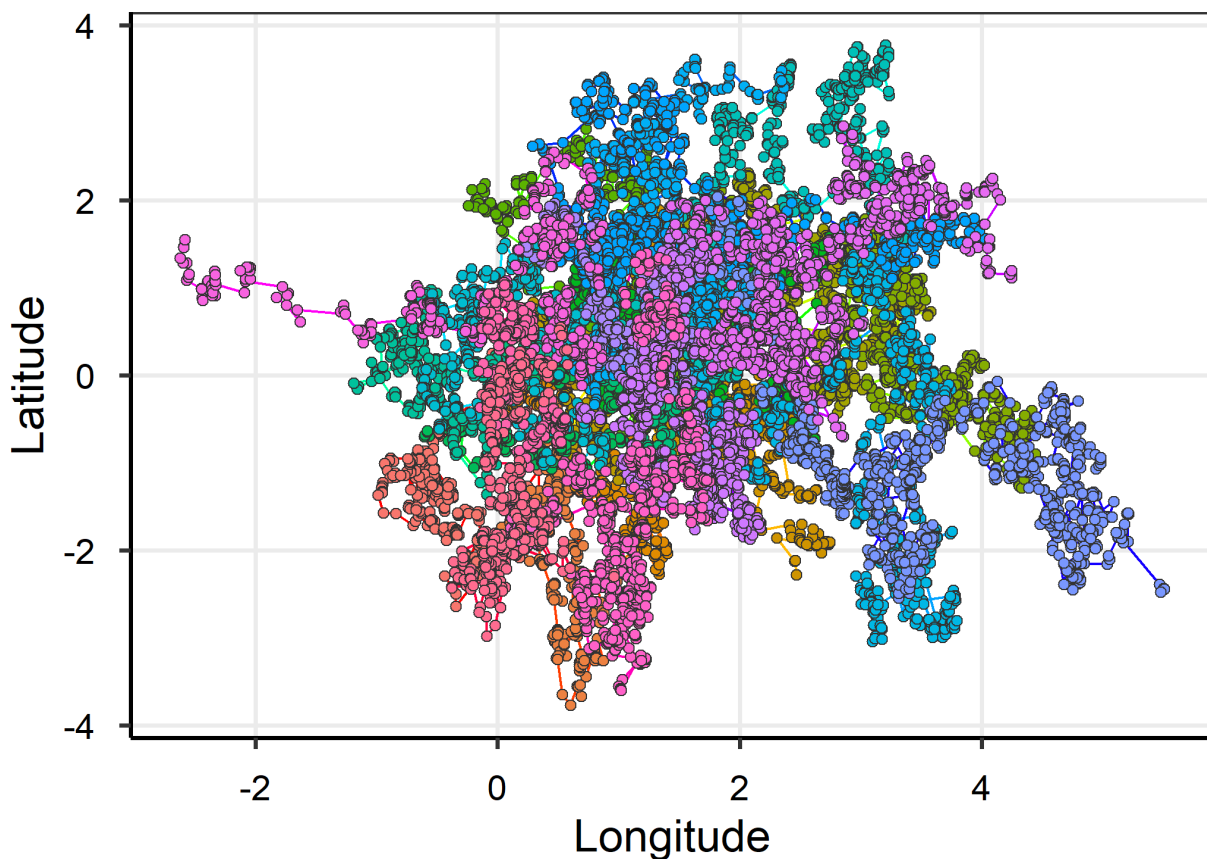


Figure V1 Map of the simulated tracking data included in the tracks dataset, created with `PlotTracks()` default settings.

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References & Recommended resources

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